

B.Tech Degree IV Semester (Supplementary) Examination February 2012

CS/EC/EB/EI 402 MICROPROCESSORS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Draw the bit pattern of Flag register of 8085 and explain the purpose of each and every flag.
(b) Explain the functions of the following pins of 8085.
(i) ALE (ii) HOLD (iii) HLDA (iv) $\overline{IO/\overline{M}}$ (v) RESET
(c) What are the different addressing modes associated with 8085 instructions? Explain with examples.
(d) Define stack and stack pointer. Explain how information is stored and retrieved from the stack.
(e) What is an interrupt? What are maskable and non maskable interrupts?
(f) Draw the timing diagram for the instruction MOV A,B (MOV A, B is stored at the location 5040H, opcode 78).
(g) Draw the block diagram of 8253/54.
(h) Briefly explain the master mode operation in DMA controller.

PART B

(4 x 15 =60)

- II. Draw the functional block diagram of 8085 and explain the internal architecture and functions of each block. (15)
- OR**
- III. (a) Explain how address bus/data bus is demultiplexed in an 8085 based system. (5)
(b) Draw and explain the register array of 8085. (5)
(c) What are buffers and latches? (5)
- IV. (a) List out various logical instructions available in 8085 and explain what happens when each of them is executed. (10)
(b) Compare the instructions CMP and SUB. (5)
- OR**
- V. Explain the classification of 8085 instruction set with example. (15)
- VI. Draw and explain the timing diagram of the instruction MVI A, 32 H. (15)
- OR**
- VII. Discuss the interrupt structure of 8085. (15)
- VIII. Explain the architecture of 8255 with neat block diagram. (15)
- OR**
- IX. Draw the functional block diagram of 8279 and explain its working. (15)